Exercises

Question 10.1: Find the magnitude of the following vectors:

(a) $\bar{a} = 4\hat{i} + 3\hat{j}$

(b) $\bar{b} = 4\hat{i} - 3\hat{j}$

(c) $\bar{c} = 3\hat{i} + 4\hat{j} + 5\hat{k}$

Question 10.2: What is the unit vector that points in the same direction as the vector $3\hat{i} + 4\hat{j} + 5\hat{k}$? A unit vector has a magnitude [length] of 1. Therefore you need to calculate the magnitude of this vector and then divide the vector by this value.

Question 10.3: Provide the vector of each of the following:

(a) The vector that represents a line from $\vec{a} = (2, -7, 0)$ to $\vec{b} = (1, -3, -5)$

(b) The vector that represents a line from $\vec{a} = (1, -3, -5)$ to $\vec{b} = (2, -7, 0)$

Question 10.4: Determine the magnitude of the following vectors:

(a) $\vec{a} = \langle 3, -5, 10 \rangle$

(b) $\vec{p} = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle$

Question 10.5: At time $t = 0$ two cars are at the origin of a set of x-y coordinates (i.e. $0 = (0,0)$). Starting at $t = 0$, car A travels along the x-axis at 30 mph. At the same time car B travels in the direction $3\hat{i} + 4\hat{j}$ with a speed of 60 mph. After 3 hours, what is the distance between the two cars in miles? This requires some thought which may be helped by drawing a diagram.

Question 10.6: Calculate the gradient, *grad* of the following scalar fields:

(a) $f(x,y) = x^2 \sin(5y)$

(b) $f(x,y,z) = z \exp(-xy)$

(c) $f(x,y) = 1 \times 10^{-3} (x - 400)^2 + 1 \times 10^{-3} y^2$

Question 10.7: For the surface / scalar field: $f(x,y) = 1 \times 10^{-3} (x - 400)^2 + 1 \times 10^{-3} y^2$, calculate the magnitude of the gradient vector for all x and y. What is the magnitude of the gradient vector at $(x,y) = (800, 400)$? For this problem you will need to calculate the partial derivatives with respect to x and y and then create a vector, before arriving at an expression for the magnitude of that vector.

Question 10.8: Find all of the first order partial derivatives for the following functions. This means we are asking you to to find all of the partial derivatives with respect to each

variable.

(a) $f(x,y) = x^4 + 6\sqrt{y} - 10$

(b) $f(x,y) = cos\left(\frac{4}{x}\right) e^{x^2y-5y^3}$

(c) $z = \frac{9u}{u^2+5v}$

Question 10.9: An equation describing a bowl shaped hill is $f(x,y) = 100\exp\left(-1\times10^{-5}\left[x^2+y^2\right]\right)$. You decide to walk along the line $y = 0$. What is the gradient $\frac{\partial f(x,y)}{\partial x}$?